Vniversitat de València Facultat de Física

Treball de Fi de Grau en Física

Training of Machine Learning Models for Stellar Characterization

Juny 2026

Alumne: Nicolás Georgakópulos Franch

Tutor: Andrés Moya Bedón

Abstract

Reliable stellar masses are required for stellar astrophysics and for the interpretation of exoplanet observations, but direct and homogeneous mass determinations are available only for limited samples. This work presents an uncertainty-aware machine-learning study of main-sequence stellar mass prediction based on a high-quality sample of 767 stars assembled mainly from eclipsing binaries, asteroseismic stars and interferometric benchmark stars. Bayesian Additive Regression Trees, a two-process heteroscedastic sparse Gaussian Process and a Hierarchical Bayesian Neural Network are trained as level-0 predictors, whose posterior predictive distributions are combined through Bayesian Hierarchical Stacking.

For FGK-type stars, using T_{eff}, ρ and [Fe/H] as input features, the stacking model reaches a mean absolute difference below 5% on a holdout test set, while showing no clear mass-dependent bias. The same framework is then applied to eclipsing-binary components using T_{eff}, L, $\log g$ and [Fe/H], in order to test whether tidal distortion affects mass predictions transferred from a purely asteroseismic training sample. For the selected sample of 85 binary components, oblateness does not produce a significant signed bias, but it is associated with an increase in the absolute mass residuals, indicating a loss of precision rather than a systematic loss of accuracy. Finally, an experiment motivated by the ESA ARIEL mission uses T_{eff}, [Fe/H] and L, excluding $\log g$ to reduce direct dependence on stellar evolutionary models, and yields mean absolute differences below 7%. The comparison with stellar-grid and linear-relation masses shows that the methods give broadly consistent but not identical results, supporting the use of probabilistic stacking as a data-driven tool for stellar-mass estimation with reduced model dependence and explicit uncertainty propagation.

Resumen

La obtención fiable de masas estelares es necesaria para la astrofísica estelar y para la interpretación de observaciones de exoplanetas, pero su determinación directa y homogénea sólo es posible mediante muestras limitadas. Este trabajo presenta un estudio mediante machine learning con tratamiento explícito de incertidumbres para la predicción de masas de estrellas de la secuencia principal, basado en una muestra de alta calidad de 767 estrellas, construida principalmente a partir de binarias eclipsantes, métodos astrosísmicos y estrellas de referencia interferométricas. Bayesian Additive Regression Trees, una Red Neuronal Bayesiana Jerárquica y un Gaussian Process disperso heterocedástico de dos procesos se entrenan como predictores de nivel 0, cuyas distribuciones predictivas posteriores se combinan mediante Bayesian Hierarchical Stacking.

Para estrellas de tipo FGK, usando T_{eff}, ρ y [Fe/H] como variables de entrada, el modelo de stacking alcanza una diferencia media absoluta inferior al 5% en un conjunto de prueba reservado a partir del dataset original, sin mostrar un sesgo signado claro dependiente de la masa. El mismo marco se aplica después a componentes de binarias eclipsantes usando T_{eff}, [Fe/H], L y $\log g$, con el objetivo de comprobar si la distorsión por mareas afecta a las predicciones de masa transferidas desde una muestra de entrenamiento puramente astrosísmica. Para la muestra seleccionada de 85 componentes de sistemas binarios, la oblaticidad no produce un sesgo signado significativo, pero sí está asociada con un aumento de los residuos absolutos de masa, lo que indica una pérdida de precisión más que una pérdida sistemática de exactitud. Finalmente, un experimento motivado por la misión ARIEL de la ESA usa T_{eff}, [Fe/H] y L, excluyendo $\log g$ para reducir la dependencia directa de modelos de evolución estelar, y obtiene diferencias medias absolutas inferiores al 7%. La comparación con masas obtenidas mediante modelos de evolución estelar y relaciones lineales muestra que los métodos proporcionan resultados ampliamente consistentes, aunque no idénticos, apoyando el uso del stacking probabilístico como herramienta basada en datos para la estimación de masas estelares con menor dependencia de modelos y propagación explícita de incertidumbres.

Contents

1 Introduction **3**

2 Machine Learning Models **4**
- 2.1 Bayesian Framework for the Models 4
- 2.2 Bayesian Additive Regression Trees (BART) 5
- 2.3 Two-process Heteroscedastic Sparse Gaussian Process (GP) 6
- 2.4 Hierarchical Bayesian Neural Network (HBNN) 7
- 2.5 Bayesian Hierarchical Stacking 8
- 2.6 Model Evaluation through Cross-validation 9

3 Methods **9**
- 3.1 Spectroscopy 9
- 3.2 Eclipsing Binaries 10
- 3.3 Asteroseismology 10
- 3.4 Interferometry 10

4 Data **11**
- 4.1 Catalogues 12
- 4.2 Cleaning 13

5 Machine Learning Model Training **15**
- 5.1 Mass Predictions for FGK-type Stars 15
- 5.2 ARIEL: Predictions with reduced model dependence 20
- 5.3 Oblateness and mass predictions of eclipsing binaries 23

6 Conclusions **28**

1 Introduction

Machine-learning methods have become increasingly important in stellar astrophysics, particularly for stellar characterization (Moya & López-Sastre, 2022). Quantities such as stellar mass, radius, effective temperature, luminosity, surface gravity and metallicity determine the physical state and evolutionary stage of a star, and they also affect the interpretation of any planetary system orbiting it. Reliable stellar parameters are therefore required. However, the direct determination of these quantities is not always possible (Serenelli et al., 2021). Different observational methods constrain different stellar parameters with different levels of model dependence, and no single method provides complete coverage across all stellar types.

For stars lacking direct mass measurements, empirical relations and numerical regression methods provide a way to infer masses from other observable stellar parameters. With the development of artificial intelligence and machine learning, more flexible methods can now be used to model these relations (Moya & López-Sastre, 2022). Their usefulness depends not only on predictive accuracy, but also on their ability to represent uncertainty and on the quality of the data from which they learn.

This work adopts a Bayesian approach to machine-learning stellar-mass prediction. Input quantities with reported uncertainties are treated as probability distributions, allowing the models to propagate observational uncertainties into their posterior predictions. This is especially relevant when the models are applied to regions of parameter space that are sparsely represented in the training data. The models used in this work, described in Section 2, are combined through Bayesian Hierarchical Stacking. This stacking approach assigns feature-dependent weights to the different predictors, allowing the final model to favour each method in the regions where it performs most reliably (Yao et al., 2022).

The resulting predictions depend strongly on the quality of the training data: precision, completeness and physical reliability of the stellar features used (Serenelli et al., 2021). For this reason, the training sample was built from reference methods that provide the most reliable available stellar parameters, mainly detached eclipsing binaries, asteroseismology and interferometry, as described in Section 3. The literature sources used to build the final catalogue were carefully selected and processed in Section 4.

The models are applied in three studies, presented in Section 5. The first application provides a baseline assessment of the predictive accuracy and bias of each method, as well as of the improvement obtained through Bayesian Hierarchical Stacking. This test is focused on FGK-type stars, which dominate the final catalogue. Its relevance to the low-mass M-star regime is indirect: in eclipsing-binary systems with FGK-type primary and a lower-mass companion, accurate characterization of the main star can provide useful constraints on the secondary Maxted (2023).

The second application is motivated by ESA's ARIEL space mission, for which accurate stellar characterization is required to interpret atmospheric observations of the exoplanets orbiting the studied stars. In this case, the mass estimates should be as independent as possible from stellar evolutionary models. The feature $\log g$ is therefore excluded because it may introduce a stronger dependence on stellar-evolution models (Tsantaki et al., 2018; Serenelli et al., 2021). The resulting predictions are also compared with masses obtained from stellar evolutionary grids and with masses derived from linear relations in Moya et al. (2018).

Finally, this work examines whether eclipsing-binary components can be treated as approximately individual stars, for the purposes of mass prediction. To test this, asteroseismic stars are taken as a reference sample of approximately isolated stellar evolution. The models are trained only on asteroseismic stars, and then applied to eclipsing-binary components. The differences between the predicted and reference masses are analysed as a function of the oblateness parameter, used here as an indicator of binary interaction. This makes it possible to assess whether mass relations calibrated on individual stars remain valid for eclipsing-binary components, or whether increasing tidal distortion produces a measurable degradation in the predictions.

2 Machine Learning Models

2.1 Bayesian Framework for the Models

Bayesian statistics treats unknown model quantities as uncertain variables. The information available before analysing the data is encoded in a prior distribution; after conditioning on the observations, this information is updated into a posterior distribution. In compact form, Bayes' theorem states that $p(\theta \mid \mathcal{D}) = p(\mathcal{D} \mid \theta)p(\theta)/p(\mathcal{D})$, where θ represents model parameters, $\mathcal{D}$ the data and $p(\mathcal{D})$ the evidence, which normalizes the posterior distribution. This is useful in scientific modelling because it makes assumptions explicit and allows uncertainty to be carried through the full inference process (McElreath, 2020).

This framework is especially relevant for stellar characterization. The training data contain observational errors, correlations between stellar parameters and regions of sparse feature coverage in parameter space. In this context, the input stellar quantities used by the models are referred to as features. A Bayesian treatment is therefore not chosen only to obtain point predictions, but to quantify how reliable those predictions are. The models discussed in the following subsections use priors to regularize flexible functions and posterior inference to propagate both data uncertainty and model uncertainty into the final estimates (Chipman et al., 2010; Rasmussen & Williams, 2006; Jospin et al., 2022).

For the models used here, the posterior distribution is usually too complex to evaluate analytically. Markov Chain Monte Carlo (MCMC) methods address this by constructing a sequence of dependent samples whose long-run distribution approximates the target posterior. A proposed move is accepted or rejected according to how compatible it is with the posterior, so the chain spends more time in high-probability regions without requiring an exhaustive search of the full parameter space (Hastings, 1970; Bardenet et al., 2017). However, because consecutive samples are correlated, practical MCMC requires convergence checks, an initial warm-up or burn-in period, and attention to the effective number of independent samples. Although computationally expensive, this sampling approach is well suited to the Bayesian models in this work because it provides a principled way to explore parameter uncertainty rather than collapsing the model to a single point estimation without error.

Rather than selecting a single best model, this work combines the predictions of independent level-0 models through Bayesian Hierarchical Stacking (BHS), as sketched in Figure 1. In the following sections, a Bayesian Additive Regression Trees, a Gaussian Process and a hierarchical Bayesian neural network model are described. The predictions yielded by these models can be passed to this first-level model, which assigns feature-dependent weights to each one, in order to optimize predictive performance.

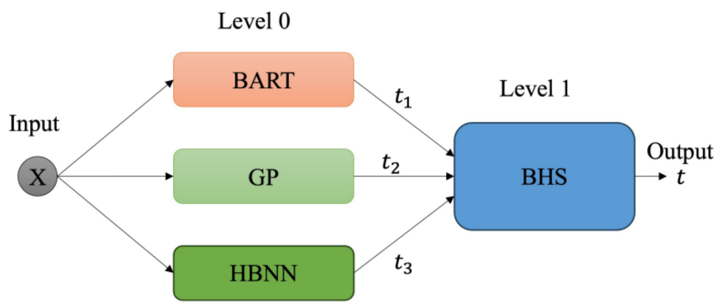


Figure 1: Schematic architecture of the Bayesian Hierarchical Stacking model. The predictions from BART, GP and HBNN are combined through a first-level model that learns their relative weights.

2.2 Bayesian Additive Regression Trees (BART)

Among the three level-0 models used in this work, the practical deployment of the BART framework is mathematically and computationally the most straightforward, since its architecture requires relatively few tuning choices (Hill et al., 2020). To understand the ensemble, the mechanics of a single regression tree must be defined. A tree recursively partitions the dataset by evaluating binary thresholds across input variables (e.g. $T_{\rm eff} < 5000\,\mathrm{K}$). At each step, the algorithm selects the threshold that maximizes the homogeneity of the target variable in the resulting subsets. When predicting for a new observation, the input traverses this learned path of decisions until it reaches a terminal node, or “leaf”. The target variable prediction associated with that leaf is simply the mean value of the training points that occupy that specific terminal node.

The Bayesian Additive Regression Trees framework, introduced by Chipman et al. (2010), employs a sum-of-trees architecture to approximate complex non-linear functions. A Random Forest constructs an ensemble of deep, independent regression trees. By training each one of such predictors on a random sample of the dataset and averaging the predictions of all of them, it produces a final output which reduces variance. BART follows this dataset subsampling logic, but the final predictions are not the average of the independent models, but the cumulative sum of the terminal node values across all interdependent trees in the ensemble. Therefore, in Random Forests every tree is a strong learner which produces an absolute value prediction for the target; while in BART the components are weak learners, which just produce a part of the prediction (Chipman et al., 2010). This weak learner approach is mathematically guaranteed by regularisation priors, which constrain the size and structural influence of each tree, preventing any single component from dominating the overall model.

Since each individual tree recursively partitions the predictor space utilising binary decision rules, the overall capacity and complexity of BART depends on the total number of trees. The model may be optimized following the Bayesian backfitting MCMC scheme introduced in Section 2.1. During each sweep, the algorithm updates one tree conditional on the partial residual left by the rest of the ensemble, proposing local changes such as growing, pruning or modifying a split. Repeating this process samples plausible tree structures and terminal-node values rather than selecting a fixed ensemble (Chipman et al., 2010).

Crucially for stellar characterization, where uncertainty of the input features is a dominant factor, the framework used in this work accommodates an errors-in-variables approach. Inputs are not treated as static, exact measurements; rather, they are modeled as latent variables characterised by a Gaussian distribution whose variance is fixed by the observational uncertainties. This allows the model to account for the diverse uncertainties present across the entire star catalogue.

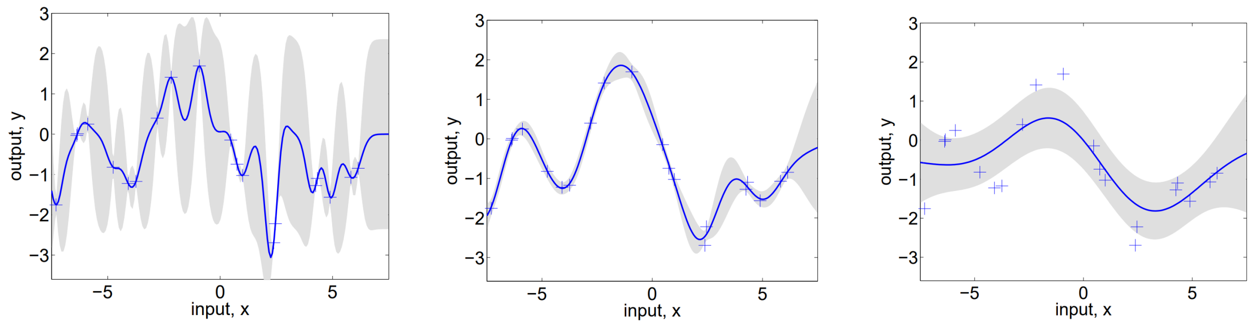
2.3 Two-process Heteroscedastic Sparse Gaussian Process (GP)

In classical parametric modelling, the functional form of the predictor is fixed in advance, and inference is performed over its parameters. A Gaussian Process (GP) takes a different approach: it defines a probability distribution over possible functions. Any finite set of function values is assumed to follow a joint Gaussian distribution, so the model is specified by a mean function and a covariance function, or kernel. In this work, the covariance between two input points is modelled with an ARD squared-expontential kernel:

$$k(\boldsymbol{x}, \boldsymbol{x}') = A^2 \exp\left[\sum_{d=1}^{D} -\frac{(x_d - x'_d)^2}{2\ \ell_d^2}\right], \tag{1}$$

where A controls the overall scale of the functions and ℓ_d is the length-scale associated with input dimension d, at points $\boldsymbol{x}$ and $\boldsymbol{x}'$. The diagonal terms describe the variance associated with each point, while the off-diagonal terms describe how strongly two different stars are correlated.

This covariance matrix is the object through which the correlations between the probability distributions of the different training points are propagated. Since the matrix has N^2 entries for N training stars, treating all of these correlations as independent quantities would place the model in a highly flexible regime, with enough freedom to reproduce noise or local fluctuations in the training data (see Figure 2a). A GP avoids this overfitting by not adjusting the covariance matrix element by element. Instead, a prior is imposed on the length-scale ℓ: nearby stars in feature space have large covariance, while the covariance decreases as the separation between their input features increases. The resulting matrix is thus condensed around the diagonal (Rasmussen & Williams, 2006).



(a) Small length-scale **(b)** Intermediate length-scale **(c)** Large length-scale

Figure 2: Example of Gaussian Process posterior predictions obtained for the same training points using different length-scales. Smaller length-scales allow the model to follow local variations more closely, while larger length-scales impose smoother correlations across the input space. The shaded regions show the corresponding predictive uncertainty. Reproduced from Figure 2.5 of Rasmussen & Williams (2006).

MIGHT DELETE THIS PRAGRAPH The version used in this work is the Automatic Relevance Determination (ARD) kernel, where each input dimension has its own length-scale rather than sharing a single one. Dimensions with little predictive information can therefore be assigned very large length-scales, making them contribute less to the covariance structure (Rasmussen & Williams, 2006).

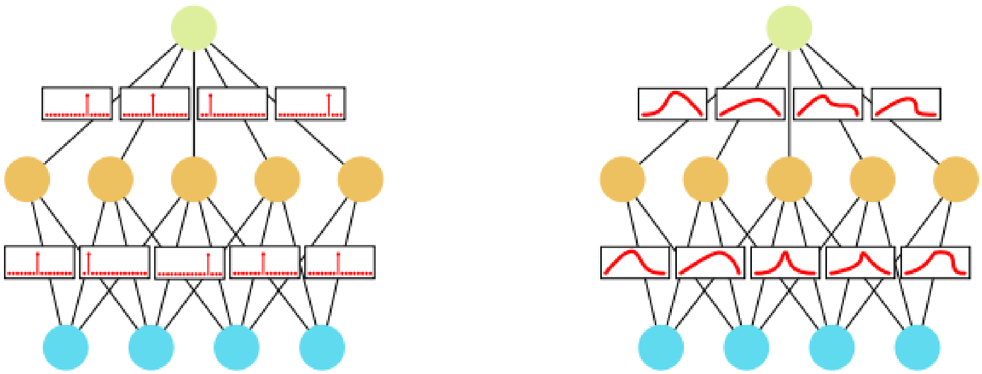
Since stellar catalogues contain measurements with uncertainties that vary between stars and between parameters, the model used in this work follows a two-process heteroscedastic construction: the predictive mean and the noise are represented by different Gaussian Processes. This allows the predictive uncertainty to vary across the input space instead of being forced into a single global noise level (Le et al., 2005). To reduce computational costs, a sparse approximation is also applied: both processes are represented through a reduced number of inducing points, which act as representatives of the entire training set. Predictive covariances are then computed through the inducing set rather than through all pairwise relations between the N observations, reducing the computational complexity (Quiñonero-Candela & Rasmussen, 2005). Even so, Gaussian Processes can remain computationally demanding for large datasets.

The model is treated in a fully Bayesian way. As described in Section 2.1, MCMC sampling explores plausible kernel hyperparameters, inducing variables and noise structures rather than fixing them at a single point with no uncertainty. Predictions are then averaged over these samples, propagating measurement and model uncertainty into the reported intervals.

2.4 Hierarchical Bayesian Neural Network (HBNN)

At its core, an artificial neural network is a sequence of mathematical operations arranged in layers. Neural networks are used to approximate non-linear relations by applying successive linear transformations and non-linear activation functions to the input data. In a standard network, the weights and biases in each layer are fitted as fixed point estimates. This can provide accurate predictions, but it does not naturally represent uncertainty in the model parameters and may produce overconfident results outside the region well covered by the training data (Jospin et al., 2022).

Bayesian Neural Networks (BNNs) address this limitation by assigning probability distributions to the weights and biases instead of single values. After conditioning on the data, the prediction is obtained by averaging over an ensemble of possible networks, making model uncertainty part of the prediction itself (Blundell et al., 2015). In the hierarchical formulation used in this work, the model also accounts for the finite precision of the observations, not only for uncertainty in the neural-network weights and biases (Jospin et al., 2022). This distinction is central for stellar characterization, where sparse training regions and measurement errors both affect the reliability of the final prediction.



(a) Point-estimate NN. **(b)** BNN with weight distributions.

Figure 3: Comparison between neural-network formulations. Figure 3a shows a standard network, in which the weights and biases are fixed after training. Figure 3b shows the Bayesian formulation used in this work, where probability distributions are assigned to the weights and biases. Adapted from (Jospin et al., 2022).

The network itself uses Gaussian priors for the weights and biases and a two-layer architecture with Leaky ReLU activations (Maas et al., 2013). As discussed in Section 2.1, MCMC sampling explores the posterior distribution rather than converging to a single point with no uncertainty. The HBNN therefore keeps the flexibility of neural networks while preserving the probabilistic interpretation required for calibrated stellar predictions (Jospin et al., 2022; Tamames-Rodero et al., 2025).

2.5 Bayesian Hierarchical Stacking

The predictions from BART, GP and HBNN are combined through Bayesian Hierarchical Stacking (BHS). Stacking is an ensemble method in which the outputs from several models are combined using learned weights, allowing relevant information from each model to contribute to the final prediction (Wolpert, 1992). Unlike simple averaging, BHS allows the weights assigned to each level-0 predictor to depend on the input features. This is important because a model may be reliable for near-solar stars but less reliable for more massive objects or for stars with larger uncertainties. The former is the case of the level-0 models used in this work, as described in Section 5. The final estimate is therefore a locally weighted combination of the three model outputs (Yao et al., 2022).

The weights are estimated using cross-validation information, discussed in Section 2.6. The hierarchical formulation regularizes the variation in these weights, preventing the stacking model from reacting too strongly to noise in sparsely populated regions of the dataset. Thus, BHS is used as a controlled way to exploit the complementary strengths of the three models while reducing the risk of relying on a single imperfect predictor.

2.6 Model Evaluation through Cross-validation

Model evaluation estimates how well a trained model will generalize to stars that were not used during training. A simple train–validation split gives a direct test of this: part of the dataset is kept aside as a validation subset, and the trained model is then evaluated against it. Nevertheless, the result can depend strongly on the particular partition, since a large validation set reduces the data available for fitting, while a small one gives a noisy estimate of the error (Rasmussen & Williams, 2006).

Cross-validation reduces this dependence on a single split. In k-fold cross-validation, the dataset is divided into k subsets; each subset is used once for validation while the remaining subsets are used for training. The reported performance is then averaged over the folds, so every star contributes both to fitting and to out-of-sample assessment (Rasmussen & Williams, 2006). This provides a more stable basis for evaluating the target-variable predictions produced by the models used in this work, allowing for a better comparison between them.

3 Methods

This section introduces the observational basis of the stellar features used later to construct the final catalogue. The catalogue was built from three reference methods: detached eclipsing binaries, asteroseismology and interferometry. Spectroscopy is discussed first as context, because it is essential for many stellar parameters, but relies strongly on stellar-atmosphere modelling. This contrast motivates the use of the other three methods as the main sources of high-quality parameters, while also making clear that they are not completely free from auxiliary assumptions.

3.1 Spectroscopy

Spectroscopy analyses the stellar radiation received as a function of wavelength. The continuum, discontinuities, absorption lines and molecular bands are produced by physical processes in the stellar atmosphere, and therefore contain information about stellar parameters.

The difficulty is that these quantities are not usually obtained by direct measurement. They are inferred by comparing the observed spectrum with synthetic spectra, line lists and stellar-atmosphere models (Tsantaki et al., 2018). Spectroscopy is therefore essential, especially for T_{eff} and [Fe/H], but it can also introduce model-dependent assumptions into any machine-learning model trained on those input features.

3.2 Eclipsing Binaries

Detached eclipsing binaries are systems in which two stars orbit each other without significant mass transfer, while their orbital plane causes periodic eclipses as seen from Earth. Because the components are dynamically linked but remain physically separated, they are especially valuable for obtaining fundamental stellar parameters such as M (Southworth, 2015; Eker et al., 2018).

The measured light curve gives the orbital period, eclipse geometry, relative radii and temperature ratio. Radial-velocity measurements from Doppler shifts then provide the orbital velocities. Combining these observables with Keplerian dynamics yields absolute masses and radii, from which density and surface gravity follow. These quantities are therefore mainly geometric and dynamical.

The atmospheric quantities are less direct. Absolute temperatures and luminosities usually require spectroscopy, colour-temperature relations, parallax, bolometric fluxes, or a combination of these ingredients (Graczyk et al., 2022). For this reason, detached eclipsing binaries are treated as a strong reference source for masses and radii, while their temperature and luminosity labels still require attention to the assumptions used to derive them.

3.3 Asteroseismology

Asteroseismology studies stellar oscillations. An extensively studied type of oscillations occurs in solar-like stars, where turbulent motion in the outer convective envelope excites pressure waves that propagate through the interior and produce small brightness variations. Time-series photometry and Fourier analysis reveal the characteristic oscillation frequencies of the star (Bellinger et al., 2019; Serenelli et al., 2021).

Two global seismic quantities are especially useful. The large frequency separation, $\Delta\nu$, scales with the square root of the mean density, while the frequency of maximum oscillation power, $\nu_{\max}$, is related to surface gravity and effective temperature:

$$\Delta\nu \propto \sqrt{\frac{M}{R^3}}, \qquad \nu_{\max} \propto \frac{M}{R^2\sqrt{T_{\mathrm{eff}}}}. \tag{2}$$

With an external estimate of T_{eff}, these relations constrain mass, radius and $\log g$ (Chaplin & Miglio, 2013).

This makes asteroseismology particularly useful for single stars, where direct dynamical masses are usually unavailable. However, the method still depends on scaling relations, external atmospheric inputs and, in more detailed analyses, stellar models. It therefore provides strong constraints, but not completely model-independent labels.

3.4 Interferometry

Interferometry combines the light collected by separated telescopes, effectively increasing the angular resolution beyond that of a single aperture. For nearby bright stars, this makes it possible to measure the angular diameter. Together with an independent distance, usually from parallax, the stellar radius can then be obtained geometrically (Ligi et al., 2016).

If the bolometric flux is also known, the effective temperature follows from the Stefan–Boltzmann relation. This makes interferometry one of the most empirical routes to stellar radii and temperatures, and it is used in benchmark-star calibrations (Soubiran et al., 2024). The method is not fully assumption-free, because bolometric flux estimates require extinction corrections and model-dependent treatment of unobserved spectral regions. It is also limited to stars with sufficiently large angular diameters, so its contribution to the total sample is small.

4 Data

In supervised machine learning, the quality of the input features can be as important as the size of the training set. Large catalogues affected by heterogeneous systematics may lead the models to reproduce those biases instead of learning the underlying physical relations (Li & Chao, 2021). For this reason, this work follows a quality-centred strategy, prioritising a smaller, traceable and physically reliable calibration set over a larger but less controlled sample.

Following the guidance of Moya et al. (2018), whose catalogue covered approximately the mass range $0.7\,M_{\odot} < M < 2.5\,M_{\odot}$, an extended stellar catalogue was assembled from literature sources with complementary parameter coverage. The lower boundary was reduced to $0.5\,M_{\odot}$, since one of the aims of this extension was to improve the coverage at lower masses. Such reference stars are needed to calibrate predictions for M-type stars, which are especially relevant in exoplanet studies (see Section 5.2). The aim was to obtain a statistically meaningful and heterogeneous sample, while preserving clear information on the uncertainties of the stellar parameters. Because the reported uncertainties are propagated into the final mass predictions in the probabilistic models in this work, this uncertainty-treatment is a key aspect.

As explained in Section 3, the reference sources were selected according to three observational methods: eclipsing binaries, asteroseismology and interferometry. Studies published between 2018 and 2026 were reviewed when they provided uncertainties and as many of the required stellar parameters as possible, listed in Table 1. Stellar mass and radius were mandatory, since stellar densities were computed from them. This initial compilation contained 16094 stars before duplicate removal and quality cleaning.

Table 1. Summary of the stellar parameters contributed by each catalogue to the final calibration sample of 767 usable main-sequence stars. Each value gives the number of stars with the corresponding available parameter. The metallicity column combines [Fe/H] and [M/H]. Catalogues not individually discussed in this work are described in Moya et al. (2018).

Catalogue	M ($M_\odot$)	R ($R_\odot$)	$T_{\rm eff}$ (K)	L ($L_\odot$)	Meta (dex)	$\log g$ (dex)	ρ ($\rho_\odot$)	a ($R_\odot$)
Serenelli et al. (2017)	240	240	240	240	240	240	240	0
Xiong et al. (2023)	141	141	141	141	141	141	141	141
Chaplin et al. (2014)	62	62	62	62	62	62	62	0
Eker et al. (2014)	54	54	54	54	54	54	54	54
Yıldız et al. (2019)	42	42	42	42	42	42	42	0
Lund et al. (2017)	28	28	28	28	28	28	28	0
Huber et al. (2013)	24	24	24	24	24	0	24	0
Silva Aguirre et al. (2015)	24	24	24	24	24	24	24	0
Wang et al. (2025)	20	20	20	20	20	20	20	20
Eker et al. (2014, 2018)	17	17	17	17	17	17	17	17
Silva Aguirre et al. (2017)	15	15	15	15	15	15	15	0
Soubiran et al. (2024)	13	13	13	13	13	13	13	0
Hełminiak et al. (2019)	11	11	11	11	11	11	11	11
Pan et al. (2025)	10	10	10	10	10	10	10	10
Ligi et al. (2016)	10	10	10	10	10	0	10	0
Chaplin et al. (2014)	8	8	8	8	8	8	8	0
Özdarcan et al. (2025)	7	7	7	7	7	7	7	7
Malkov et al. (2007)	4	4	4	4	4	0	4	0
Welsh et al. (2012)	4	4	4	4	4	4	4	4
Brogaard et al. (2018)	3	3	3	3	3	3	3	3
Özdarcan et al. (2024)	3	3	3	3	3	3	3	3
Torres et al. (2010)	3	3	3	3	3	3	3	3
Çakırlı et al. (2025a)	2	2	2	2	2	2	2	2
Çakırlı et al. (2025b)	2	2	2	2	2	2	2	2
Knudstrup et al. (2020)	2	2	2	2	2	2	2	2
Martín-Ravelo et al. (2024)	2	2	2	2	2	2	2	2
Miller et al. (2022)	2	2	2	2	2	2	2	2
Miller et al. (2026)	2	2	2	2	2	2	2	2
O'Brien et al. (2025)	2	2	2	2	2	2	2	2
Xu et al. (2026)	2	2	2	2	2	2	2	2
Yücel et al. (2026)	2	2	2	2	2	2	2	2
Bond et al. (2020)	1	1	1	1	1	1	1	1
Jennings et al. (2024)	1	1	1	1	1	1	1	0
Lund (2019)	1	1	1	1	1	1	1	0
Maxted (2023)	1	1	1	1	1	1	1	1
Maxted (2025)	1	1	1	1	1	1	1	1
Thomsen et al. (2022)	1	1	1	1	1	1	1	1
Total	767	767	767	767	767	729	767	295

4.1 Catalogues

The literature sources used to compile the catalogue are summarised in Table 2. The main role of each source in the construction of the final sample, is also indicated, since not all catalogues contributed the same type of information.

Table 2. Summary of the main literature sources used to build the stellar catalogue. The initial-entries column gives the number of main-sequence stars taken from each source before duplicate removal and further quality cleaning (see Section 4.2). The method column indicates the main observational basis of each source, while the final column summarises its role in the catalogue construction and any relevant technical treatment applied during the merging process.

Source	Initial entries	Main method	Main contribution	Notes for this work
Eker et al. (2018)	508	Detached eclipsing binaries	Accurate stellar parameters for main-sequence binary components over a broad mass range	Used as a high-quality reference for masses and radii; also relevant for calibrating L and $T_{\rm eff}$.
Hełminiak et al. (2019)	22	Detached eclipsing binaries with spectroscopic support	Binary-star parameters with additional atmospheric information	Treated with care because the reported abundance information was given as [α/Fe], not directly in the metallicity scale adopted here.
Graczyk et al. (2021, 2022)	28	Detached eclipsing binaries	Surface-brightness–colour calibration samples	The dwarf-star calibrating sample of Graczyk et al. (2022) was adopted as a strong reference source.
Nsamba et al. (2025)	8	Benchmark stars	Pre-existing benchmark-star compilation	Required careful duplicate removal because of overlap with other catalogues.
Bellinger et al. (2019)	97	Asteroseismology	Asteroseismic masses and radii for mostly solar-like stars	Several objects lacked some atmospheric parameters required in this work.
Serenelli et al. (2021)	106	Mixed benchmark compilation	Stellar mass-determination methods across the HR diagram, including eclipsing binaries and asteroseismology	Contributed a large number of reliable stars, although luminosity and metallicity were not always available.
Soubiran et al. (2024)	31	Interferometry and benchmark-star calibration	Gaia FGK benchmark stars with fundamental $T_{\rm eff}$ and $\log g$	Luminosities were provided as $\log L$, requiring conversion before use in the catalogue.
Xiong et al. (2023)	354	Eclipsing binaries with spectroscopic information	Empirical library based on binary-star parameters	Treated with additional caution because of non-standard identifiers and some comparatively large uncertainties.
Sayeed et al. (2025)	694	Asteroseismology	Homogeneous catalogue of Kepler main-sequence and subgiant solar-like oscillators	Metallicity was not always accompanied by uncertainty.
Yıldız et al. (2019)	74	Asteroseismology	Masses and radii for Kepler and CoRoT solar-like oscillators	Metallicities reported as surface metal mass fractions were converted to the logarithmic scale.

Yıldız et al. (2019) provided metallicities as surface metal mass fractions Z_s. They were converted to the logarithmic scale using $[\mathrm{M/H}] = \log\left[(Z_s/X_s)/(Z/X)_\odot\right]$, adopting $X_s \simeq 0.7381$ and $(Z/X)_\odot = 0.0181$ from Asplund et al. (2009).

4.2 Cleaning

When the same star appeared in several catalogues, the retained entry was chosen according to the reliability of the source. Priority was given to catalogues providing the most direct stellar parameters, smaller reported uncertainties and a more complete set of parameters. Regularly updated sources and benchmark-star compilations were also preferred over more heterogeneous or less traceable ones. This meant favouring high-precision eclipsing-binary catalogues such as Eker et al. (2018) and the calibrating samples of Graczyk et al. (2022), benchmark compilations such as the Gaia FGK benchmark-star catalogue (Soubiran et al., 2024), and samples cross-validated across different methods, such as those in Serenelli et al. (2021). After duplicate removal, 13783 stars remained.

The cleaned sample was then restricted to main-sequence candidates using a criterion based on $\log g$. Following Bilir et al. (2008), stars with $\log g > 4.0$ were classified as main-sequence objects, except for extremely large values of $\log g$ ($\gtrsim 5.5$), which were not interpreted as normal main-sequence stars. Although this threshold is approximate, the effect of applying different classification criteria was already analysed by Moya et al. (2018). They found that the choice between a simple threshold and a more robust classification method had little influence on the final results. The $\log g$ criterion adopted here was therefore considered sufficient for the purposes of this work.

Given the large number of red giants, only 1915 stars remained after this selection, although not all of them contained every required parameter. In principle, incomplete entries could be retained in a Bayesian framework by treating missing quantities as latent variables, provided that an explicit model for the missing-data process is specified (McElreath, 2020, p. 511). In this work, however, parameter completeness and direct traceability were prioritised, so this kind of imputation was not used.

The newly compiled 1915-star sample was then cross-matched with the catalogue of Moya et al. (2018) in order to merge both sources, while avoiding duplicates and retaining only the highest-quality entries. Following the same data-centred strategy, stars were removed when they had excessively large uncertainties, zero reported uncertainty in any parameter except a, were not classified as main-sequence stars in both catalogues, or had duplicated identifiers. This reduced the sample to 784 stars. Section 5.1 explains the later removal of 17 additional objects, leading to the final training sample of $n = 767$ stars.

Although each star in the final merged sample is assigned a primary catalogue identifier according to the reliability criteria described above, the individual stellar parameters for a given entry do not always originate from that single source. To maximise completeness, the merging algorithm constructed composite entries: when the primary catalogue lacked a given parameter, the missing value was filled from an available secondary source. The final parameter set for a star can therefore be heterogeneous in origin. To preserve traceability, a more detailed internal provenance table was used alongside Table 1; for each primary source, it records which secondary catalogues were queried to fill each missing parameter. When several secondary sources were available, they were prioritised according to the reliability hierarchy described above.

The Hertzsprung–Russell diagram was used as a final consistency check for the cleaned sample. This diagram relates stellar luminosity to effective temperature and is commonly used to identify evolutionary stages, especially the main sequence. As shown in Figure 4, the stars mostly follow the expected main-sequence behaviour, with luminosity decreasing towards lower effective temperatures. FGK stars occupy approximately $T_{\rm eff} \sim 3500$–$7400\,\mathrm{K}$ (Pecaut & Mamajek, 2013), or $\log_{10} T_{\rm eff} \sim 3.54$–$3.87$. Most of the sample lies in this regime, especially the asteroseismic and interferometric stars. The eclipsing-binary subsample is also mainly concentrated around the main sequence, but it spans a broader region of the diagram.

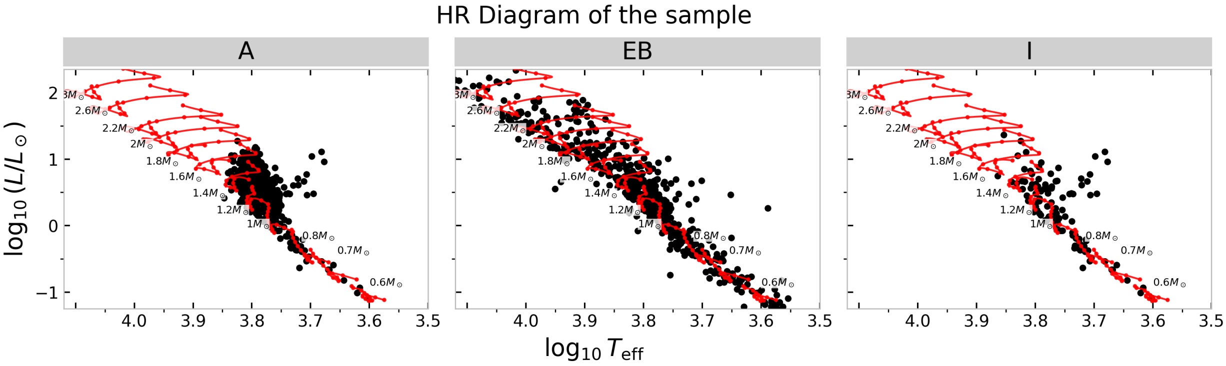


Figure 4: HR diagram of the 767 stars in the final sample. The red curves are theoretical evolutionary tracks of different masses obtained with PARSEC (Bressan et al., 2012). The sample is divided by the main method used to study each star: asteroseismology, eclipsing binaries and interferometry.

Finally, Section 5.3 requires the computation of oblateness (Equation 4), which depends on the orbital semi-major axis a. This parameter was often reported without an uncertainty. To retain a statistically meaningful sample, the missing uncertainty was estimated from four significant digits of the reported value. This produced 295 stars with computed oblateness, 48 of which used this uncertainty estimate.

5 Machine Learning Model Training

The behaviours and quantitative results described in these sections were consistent across different random-seed runs for the four analysed models, keeping the same inputs, target and training strategy. The main evaluation metrics are the Mean Absolute Error (MAE), which measures the average absolute difference between the true masses m_i and the model estimates $\hat{m}_i$; the Mean Absolute Relative Difference (MARD), which measures the average relative accuracy across N samples; and the Mean Relative Difference (MRD), which measures the signed relative bias of the predictions with respect to the true values.

$$\mathrm{MAE} = \frac{\sum_{i=1}^{N} |m_i - \hat{m}_i|}{N}, \quad \mathrm{MARD} = \frac{\sum_{i=1}^{N} \frac{|m_i - \hat{m}_i|}{m_i}}{N}, \quad \mathrm{MRD} = \frac{\sum_{i=1}^{N} \frac{m_i - \hat{m}_i}{m_i}}{N}. \tag{3}$$

5.1 Mass Predictions for FGK-type Stars

Main-sequence stars of spectral types F, G and K are especially useful in stellar studies because they are common survey targets and therefore often have well-measured atmospheric parameters. By contrast, main-sequence M-type stars are less extensively documented. They are cooler, low-mass objects that burn hydrogen slowly and evolve very little over the age of the Universe (Laughlin et al., 1997). Their intrinsic faintness and slow evolution make precise stellar parameters difficult to obtain (Chabrier & Baraffe, 2000).

Maxted (2023) proposed a way to obtain information about M-type stars by studying eclipsing-binary systems in which the M star is the secondary component of a better-characterized primary star. Because many eclipsing-binary parameters are measured relative to the two components, information about one star can help constrain the other (Southworth, 2015). This motivates a focus on systems with an FGK-type primary and an M-type secondary: accurate mass predictions for the better-characterized FGK component can improve the future calibration of the lower-mass companion. This would help to extend the calibration towards the M-star regime, where high-quality masses remain scarce.

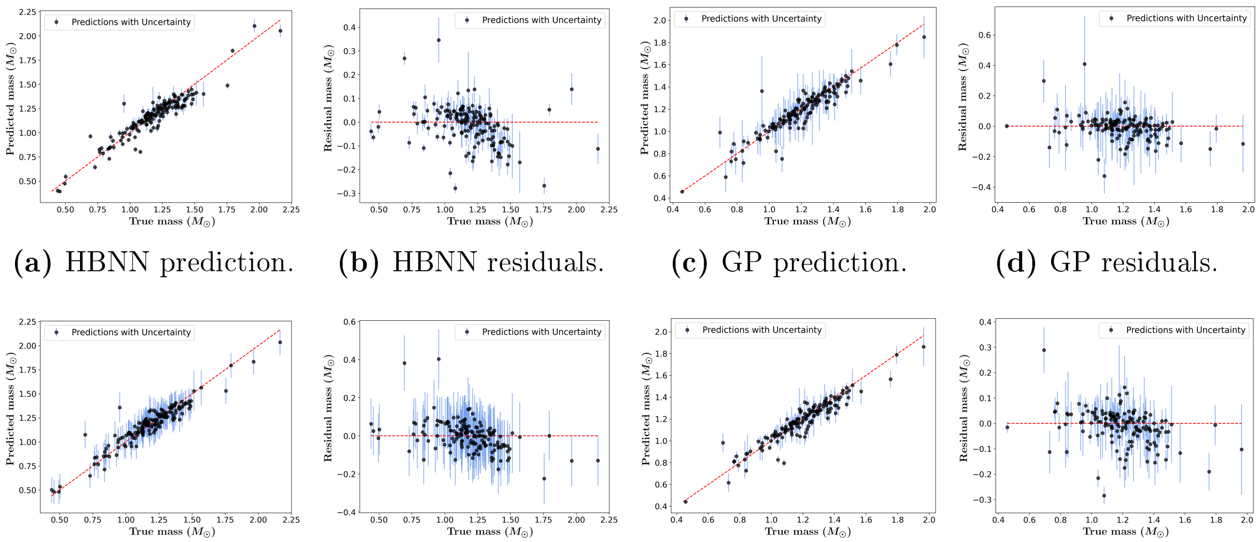
As discussed in Section 4.2, Figure 4 shows that the catalogue compiled in this work is rich in FGK-type stars, making this first mass-prediction experiment well suited to the available data. Following Moya et al. (2018), $T_{\rm eff}$, ρ and [Fe/H] were adopted as input parameters for stellar-mass estimation. This choice is motivated by well-studied eclipsing binaries, for which these quantities can be obtained more precisely than alternatives such as $\log g$ or L. After combining the stars compiled in this work with the Moya catalogue, applying the main-sequence selection described in Section 4, requiring these three input parameters, and removing stars with zero reported uncertainty, the final available set contained 767 stars. Of these, 614 were used for training and the remaining 153 stars, corresponding to 20%, were retained as a holdout set for testing. The holdout set was repeatedly randomly selected for each seed run.

Results

Figure 5a shows the HBNN mass predictions for one representative run. In the prediction panel, the points mostly follow the 1:1 relation between predicted and true mass. However, the distribution is not completely uniform: lower-mass stars tend to be overpredicted, whereas higher-mass stars tend to be underpredicted. When averaged over the full holdout set, these two effects partly cancel, leading to a Mean Relative Difference close to zero, 0.71%, as shown in Table 3.

This trend is more clearly visible in the residual plot[1], Figure 5b. The residuals change sign at approximately $1.3\,M_{\odot}$: lower-mass stars tend to have positive residuals, while higher-mass stars tend to have negative residuals. Thus, HBNN performs well for the main, near-solar-mass population, where most predictions agree with $M_{\rm true}$ within 1σ. The model should, however, be treated with caution outside this central region, especially for higher masses.

[1]For visual clarity, residuals were defined as $M_{\rm pred} - M_{\rm true}$ in the figures, including 5, while in the MAE, MARD and MRD definitions the sign is switched, $M_{\rm true} - M_{\rm pred}$.



(a) HBNN prediction. **(b)** HBNN residuals. **(c)** GP prediction. **(d)** GP residuals.

(e) BART prediction. **(f)** BART residuals. **(g)** BHS prediction. **(h)** BHS residuals.

Figure 5: Holdout-test performance of the HBNN, GP, BART and BHS mass models trained with $T_{\rm eff}$, ρ and [Fe/H]. For each model, the prediction panel shows $M_{\rm pred}$ as a function of $M_{\rm true}$, while the residual panel shows $M_{\rm pred} - M_{\rm true}$ as a function of $M_{\rm true}$. Vertical error bars represent the predictive uncertainty returned by each model. The GP and BHS panels are restricted to the stars below the 95th percentile of predicted mass uncertainty in order to keep the main population visually readable. The full calibration sample contains $n = 767$ stars, with 20% reserved for the holdout test.

A plausible explanation is that the models were trained on a sample strongly concentrated around $1\,M_\odot$. As a result, HBNN tends to pull predictions towards this well-populated region, producing masses closer to the central training distribution than to the true values at the edges of the mass range. This behaviour is expected for Bayesian neural networks when the test inputs differ from the training distribution (Izmailov et al., 2021).

The trend is less clear for the Gaussian Process model, as shown in Figure 5c. For visualization, the GP plots were restricted to stars with predicted mass uncertainty below the 95th percentile, because the full GP output contains a small number of objects with extremely large predicted uncertainties. Including those points compresses the vertical scale and makes the behaviour of the bulk of the sample unreadable.

Most GP predictions remain close to the 1:1 relation, and almost all points are compatible with the reference masses within 1σ. The residual plot in Figure 5d shows a more compact and symmetric distribution around zero than the HBNN residuals in Figure 5b. This is also reflected in Table 3: the GP reaches a MARD of 4.79%, lower than the HBNN value of 5.27%. In addition, the GP has an MRD close to zero, 0.34%, indicating no clear global tendency to underpredict or overpredict stellar masses.

It is, however, notable that the GP uncertainties are larger than in the HBNN case, indicating that the model is underconfident in part of the sample. This remains true even after restricting the plotted set to the stars below the 95th percentile of predicted mass uncertainty. Stars with input-feature uncertainties larger than 9% generated predicted mass uncertainties as large as $10^3\,M_\odot$, so they were removed from the sample, in order to investigate the origin of this extremely underconfident behaviour. This removed 17 objects, but did not eliminate the problem.

In sparse regions of the input space, Gaussian Processes naturally return broader predictive distributions, reflecting extrapolation uncertainty (Cho et al., 2026). This is one of the main advantages of GPs: predictive uncertainty increases when the model has little nearby information. This explains why no strong systematic trend is visible in Figures 5c and 5d. However, the very large uncertainty values found here, sometimes of order 10^2–$10^3\,M_\odot$, are not physically meaningful.

A further inspection of the objects with very large predicted uncertainties, after removing those stars with input parameter uncertainties larger than 9%, showed that these input objects were no longer the same across different runs, and no physical pattern was found in their stellar parameters. This suggests that the issue is not only caused by a few erroneous data entries, but is likely connected to instability in the GP posterior or in the variance submodel. Le et al. (2005) discuss GP regression with input-dependent noise and note that directly estimating both the mean and the variance leads to difficult optimisation problems. For this reason, the GP results should be interpreted with caution. The percentile-restricted plots are used only to display the behaviour of the main population, not to hide the existence of these unstable uncertainty estimates.

For BART, Figures 5e and 5f, the result is similar to that obtained with the Bayesian neural-network model: masses below $1\,M_\odot$ tend to be overpredicted, while masses above $1\,M_\odot$ tend to be underpredicted. The predictions remain relatively close to the 1:1 relation, yielding a low MARD value of 4.98%. The negative MRD value, -0.63%, shows that low-mass overpredictions and high-mass underpredictions again partially cancel when averaged across the full data interval. Therefore, as in the HBNN case, the MRD alone is not sufficient to describe the model behaviour unless it is interpreted together with the prediction and residual plots.

Here, the reduced accuracy outside the central training region is consistent with a known limitation of tree-based regression models. Standard BART and related tree ensembles are strong interpolators, but their extrapolation ability is limited by their leaf-based, piecewise-constant structure. Predictions outside sparsely sampled regions can become biased towards the dominant range of the training data and may have poorly calibrated uncertainty (Zhang et al., 2019). BART also shows comparatively large uncertainties, so the loss of accuracy away from the central mass range is reflected in underconfident predictions.

Finally, the stacking model was applied to combine the information contained in the individual BART, GP and HBNN predictions. Since both HBNN and BART show systematic behaviour away from the region near $1\,M_\odot$, the GP model is expected to have a larger influence in those outer regions. Therefore, the BHS plots shown in Figures 5g and 5h are also restricted to stars below the 95th percentile of predicted mass uncertainty, avoiding the visual masking produced by a small number of extremely uncertain points.

Figure 5g shows no clear mass-dependent trend in the BHS predictions. Therefore, the systematic mispredictions observed for BART and HBNN away from one solar mass are not inherited by the stacking model. The resulting MRD value, 0.29%, is close to zero and is the lowest among the four models, supporting the absence of a visible global bias. In this case, the behaviour of the plots confirms that the near-zero MRD is not mainly produced by localized overpredictions and underpredictions cancelling each other out, as happened for HBNN and BART.

Although the GP model clearly influences BHS, since the negative-slope trend is no longer visible, the difference between the BHS MARD, 4.66%, and the GP MARD, 4.79%, shows that the stacking model is not simply mimicking the GP model overall. Instead, it combines the models in a way that improves the final residual metrics while reducing the systematic behaviour seen in some of the individual predictors.

The predictive uncertainties observed in the plots of the BHS model are, however, the largest among the models in this representative run. This is consistent with the fact that stacking combines predictive distributions from all models and can inherit part of the large uncertainties that both the GP and BART components show for some stars (Yao et al., 2022).

Table 3. Comparison of the residual metrics obtained by the mass-prediction models on the holdout set. The models were trained with $T_{\rm eff}$, ρ and [Fe/H] as input features, and stellar mass as the target.

Metric	BART	GP	HBNN	BHS
MAE ($M_\odot$)	0.057	0.053	0.060	0.052
MARD [%]	4.98	4.79	5.27	4.66
MRD [%]	−0.63	0.34	0.71	0.29

The evaluation metrics (Table 3) show that, overall, the relative errors are consistently around the 5% level, which is a strong result for this calibration sample. Similar values were obtained by the BHS model presented by Moya & López-Sastre (2022). However, that model used four input features instead of three, and would therefore have access to more predictive information, as discussed by those authors. The 2022 model did not propagate the observational uncertainties of the stellar parameters. The fact that the present three-feature models reach comparable MARD values while incorporating these uncertainties suggests that the improved probabilistic treatment provides a more robust calibration withour a clear loss of predictive accuracy.

This interpretation is also supported by the MAE comparison. Moya & López-Sastre (2022) reported a stacking-model MAE of $0.048\,M_\odot$, which is very close to the BHS value of $0.052\,M_\odot$ obtained here. Considering the remaining models presented by Moya & López-Sastre (2022), with MAE values in the range 0.06–0.08 $M_\odot$, the models in this work achieve lower absolute errors in all cases, while treating the reported input uncertainties consistently.

Finally, all MRD values are below 1%. For GP and BHS, this is consistent with the absence of a clear global bias in the prediction plots. For HBNN and BART, however, the near-zero MRD should not be interpreted alone: in those cases, it is mainly a consequence of systematic overpredictions and underpredictions cancelling when averaged over the full holdout set.

5.2 ARIEL: Predictions with reduced model dependence

ARIEL, the Atmospheric Remote-sensing Infrared Exoplanet Large-survey, is ESA's fourth medium-class mission, M4, within the Cosmic Vision programme, and is planned for launch around 2030. It is dedicated to the atmospheric characterization of exoplanets and will study the chemical composition of the atmospheres of approximately 1000 transiting planets. The mission is based on spectroscopic and photometric observations, where the planetary signal is measured relative to the emission of the host star. Accurate and homogeneous stellar characterisation is therefore essential for the interpretation of ARIEL observations.

Within this context, the Age, Mass and Radius sub-working group aims to provide homogeneous estimates of stellar ages, masses and radii for the ARIEL target stars (Ariel Stellar Characterisation Working Group, n.d.). The Astronomy and Astrophysics Department at the Universitat de València manages this stellar characterization working group. The task is approached through complementary implementations, including both stellar-modelling methods and data-driven ML/AI techniques. The former implementation is supported by the Astrophysics and Space Sciences Institute of the Universidade do Porto.

Surface gravity, g, is a physically informative quantity for the estimation of stellar masses and radii. Previous empirical and machine-learning studies have shown that its inclusion can improve predictive performance (Moya et al., 2018). However, g is often determined through procedures that depend on stellar models. Using it as an input feature in the data-driven model would therefore reduce the independence between the two complementary approaches considered in the ARIEL stellar-characterization work. For this reason, the models in this section were trained using T_{eff}, [Fe/H] and L as input features. These quantities are not completely free from methodological assumptions, but they avoid the same direct dependence on stellar evolutionary models.

The procedure described in Section 5.1 was then repeated using this updated set of input features, yielding similar results, both visually and quantitatively. In this case, however, an additional comparison was possible. The Institute of Astrophysics of the Universidade do Porto provided a complementary dataset of stellar masses obtained through stellar-modelling methods. These estimates are based on PARSEC, the PAdova and TRieste Stellar Evolution Code, which provides stellar evolutionary tracks and isochrones used to infer stellar properties from theoretical models (Bressan et al., 2012). This dataset was first cleaned to remove stars that had already been used during training, and was then used to obtain mass predictions with the trained models. The new mass predictions, including their uncertainties, were compared with the masses obtained from stellar evolutionary grids. Linear relations developed by Moya et al. (2018) were added to the comparison as well. This provides a benchmark against both model-based mass estimates and previously published data-driven mass-estimation methods.

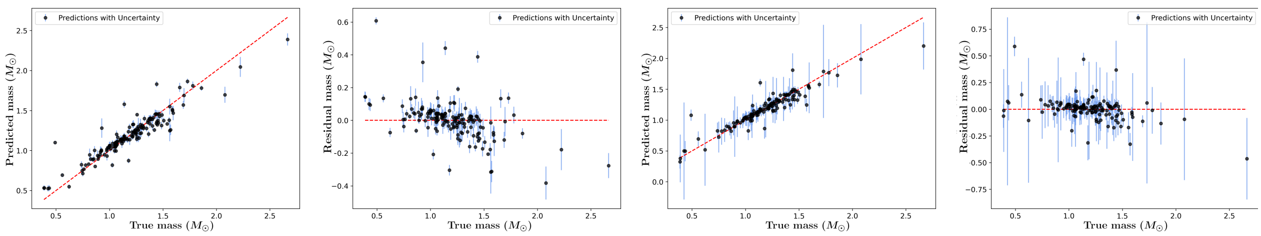
Results

The evaluations against the 20% holdout test were performed using the MAE, MARD and MRD indicators introduced in Section 5.1. Table 4 shows that the behaviour of the models is consistent with that found when training with ρ instead of L, although the performance is slightly worse, with MARD values increasing by about two percentage points. This supports the conclusion of Moya et al. (2018) that ρ is a particularly informative parameter for stellar-mass prediction. Nevertheless, Figure 6 puts this loss of accuracy into context, since the points remain only moderately dispersed around the 1:1 relation. Even with this decrease in performance, the results remain acceptable for the purposes of this comparison; the MAE values obtained, all within the range 0.07–0.08 $M_\odot$, are comparable to those reported for different models by Moya & López-Sastre (2022), which lie within the range 0.06–0.08 $M_\odot$.

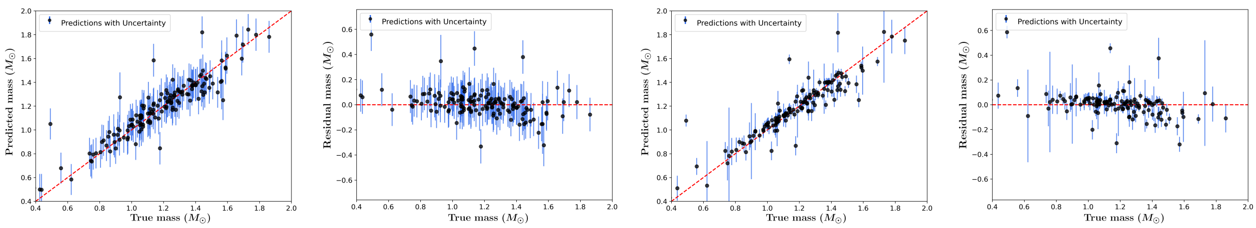
Table 4. Comparison of the residual metrics obtained by the mass-prediction models on the holdout set. The models were trained with T_{eff}, L and [Fe/H] as input features, and stellar mass as the target.

Metric	BART	GP	HBNN	BHS
MAE ($M_\odot$)	0.076	0.077	0.074	0.072
MARD [%]	6.99	6.79	7.02	6.53
MRD [%]	−2.39	−1.14	−1.89	−1.36

BART and HBNN again show the strongest systematic residual structure, with a tendency to drive predictions towards the central region of the training distribution, while GP yields the lowest MRD but retains the uncertainty-instability already discussed in Section 5.1.



(a) HBNN prediction. **(b)** HBNN residuals. **(c)** GP prediction. **(d)** GP residuals.



(e) BART prediction. **(f)** BART residuals. **(g)** BHS prediction. **(h)** BHS residuals.

Figure 6: Holdout-test performance of the HBNN, GP, BART and BHS mass models trained with T_{eff}, L and [Fe/H]. For each model, the prediction panel shows M_{pred} as a function of M_{true}, while the residual panel shows $M_{\mathrm{pred}} - M_{\mathrm{true}}$ as a function of M_{true}. Vertical error bars represent the predictive uncertainty returned by each model. The GP and BHS panels are restricted to the stars below the 90th percentile of predicted mass uncertainty in order to keep the main population visually readable. The full calibration sample contains $n = 767$ stars, with 20% reserved for the holdout test.

BHS combines the behaviours of the three level-0 models and remains the most stable compromise between residual accuracy and global bias, since its predictions remain well concentrated around the 1:1 relation, with no clear systematic trend. Since GP performs better away from $1\,M_\odot$ than BART and HBNN, the stacking approach is expected to assign it larger weights, although the reported MRD is slightly worse for BHS than for GP. This is expected because the stacking model does not simply reproduce the GP predictions, but balances them with the contributions of BART and HBNN. The large individual predictive uncertainties of BHS also suggest that it inherits part of the underconfidence already present in BART and GP, a behaviour also observed in the ρ-training case.

The mass predictions, including their uncertainties, were then compared with the stellar evolutionary-grid estimates based on PARSEC (Bressan et al., 2012) and with the linear-relation masses presented by Moya et al. (2018). The differences were analysed through fractional residuals, defined as $(M_1 - M_2)/M_2$, where M_1 and M_2 correspond to the first and second methods indicated in each comparison label in Figure 7. The closest agreement is found between the BHS predictions and the grid-model masses: in this case, all residuals remain within approximately $\pm 10\%$. The stacking approach gives systematically higher masses than the grid models, but in a relatively homogeneous way, since both the mean and the median lie close to each other, near a fractional difference of $+2\%$.

When BHS is compared with the Moya et al. (2018) linear relations, the behaviour changes, showing a more dispersed distribution with a tail of mass differences up to about 15%. The near-zero mean residual should not be interpreted as evidence that the stacking approach and the linear relations are consistently equivalent, but rather as the result of a broader residual distribution in which positive and negative deviations partly cancel. Table 5 confirms this interpretation: the mean residual is -0.38%, while the mean absolute residual increases to 3.14%, larger than the BHS-grid comparison value of 2.86%. The two methods therefore provide different predictions for individual stars, even though their average signed offset is small.

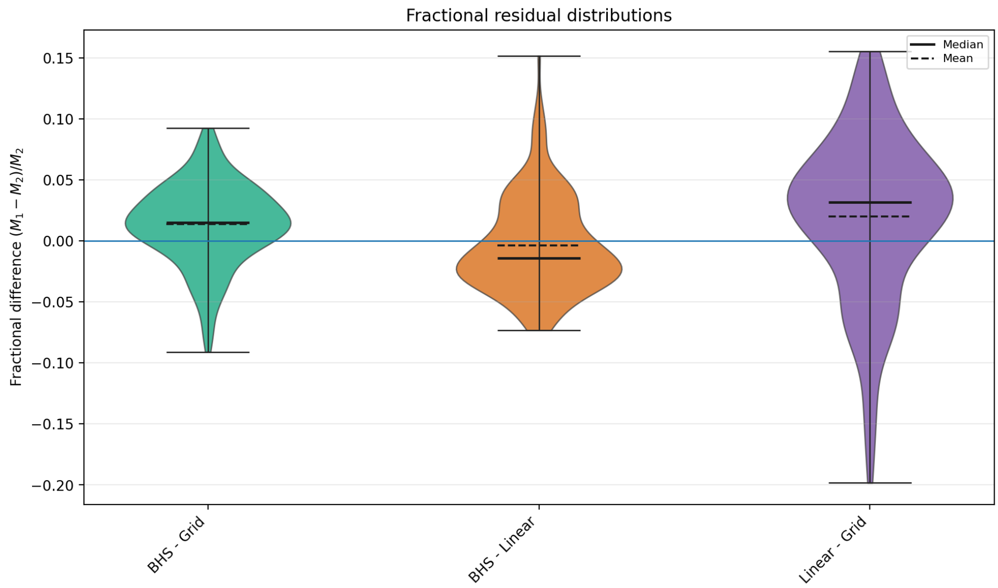


Figure 7: Distributions of fractional mass residuals for the comparison between the Bayesian Hierarchical Stacking predictions and the two reference mass estimates used for the ARIEL sample. The green distribution compares BHS with the stellar-model grid masses provided by the IA-Porto group, while the orange distribution compares BHS with the linear-relation masses from Moya et al. (2018). The purple distribution shows the direct comparison between the linear-relation and grid masses. The fractional residuals are defined as $(M_1 - M_2)/M_2$, following the order of the two methods shown on the horizontal axis. The sample contains $n = 184$ stars.

The two reference models do not produce equivalent predictions either. Figure 7 and Table 5 show that the linear-relation and grid masses differ more strongly from each other than BHS differs from either of them, with a MARD above 5% and a broad residual distribution that includes a tail of large negative residuals extending almost to -20%. This tail is consistent with the one observed in the BHS-linear comparison, suggesting that the linear relations produce comparatively low masses for a small number of stars.

Table 5. Comparison of the residual metrics obtained for the ARIEL mass estimates. The columns compare the BHS predictions with the Bressan et al. (2012) stellar evolutionary-grid masses, the Moya et al. (2018) linear-relation masses, and the linear-relation masses with the grid masses, using the same fractional-residual convention as in Figure 7.

Metric	BHS - Grid	BHS - Linear	Linear - Grid
MAE ($M_\odot$)	0.031	0.035	0.061
MARD [%]	2.86	3.14	5.60
MRD [%]	1.39	-0.38	2.01

5.3 Oblateness and mass predictions of eclipsing binaries

As discussed in Section 5.1, one possible application of accurate FGK-star mass predictions is the study of eclipsing-binary systems containing a well-characterized FGK-type component and a lower-mass companion. Before such predictions can be used in that context, it is necessary to test whether eclipsing-binary components can be treated as approximately individual stars for the purposes of mass prediction.

Eclipsing-binary components are not completely independent objects. Their mutual gravitational interaction can modify their morphology through tidal forces, preventing them from remaining perfectly spherical. This distortion can be quantified through the oblateness parameter o, given by Equation (43) of Chandrasekhar (1933), which classifies stars from perfectly spherical ($o = 0$) to highly oblate ($o \sim 1$):

$$o = 0.6446 \cdot \left(1 + 4\frac{m}{m_\star}\right) \cdot \left(\frac{R_\star}{a}\right)^3, \tag{4}$$

where $R_\star$ and $m_\star$ are the radius and mass of the star whose oblateness is being computed, m is the mass of its companion, and a is the orbital semi-major axis.

To test whether this tidal distortion affects the mass predictions, the models were trained on asteroseismic stars, which are treated here as the single-star reference sample. Predictions were then obtained for the eclipsing-binary components available in the catalogue. The predicted masses were compared with the reference masses of the eclipsing-binary stars, and the resulting residuals were analysed as a function of oblateness. This follows the general approach of Maxted (2023), where eclipsing binaries are used to evaluate how well single-star relations can be transferred to binary systems.

Results: holdout test

The models were trained using the asteroseismic stars from the dataset in Table 1. Since the goal in this section is to obtain accurate mass predictions before analysing the residual dependence on oblateness, four input features were used, following the recommendation of Moya & López-Sastre (2022): $T_{\rm eff}$, $\log g$, L and [Fe/H]. This subset contains 420 stars, of which 84 were retained as the holdout set. There are therefore two competing effects on the predictive performance of the models. On the one hand, the models now use four input features instead of three, which should improve the precision of the predictions. On the other hand, the available sample is reduced from the original 767 stars to 420 stars.

Table 6. Comparison of the residual metrics obtained by the mass-prediction models on the holdout set. The models were trained with $T_{\rm eff}$, $\log g$, L and [Fe/H] as input features, and stellar mass as the target.

Metric	BART	GP	HBNN	BHS
MAE ($M_\odot$)	0.052	0.042	0.045	0.043
MARD [%]	4.50	3.84	3.98	3.85
MRD [%]	−0.99	−0.69	−0.61	−0.65

The predictions and residuals behave similarly to those discussed in Figure 5. HBNN and BART provide systematic mispredictions away from $M = 1\,M_\odot$, while the characteristic GP behaviour, with a small number of objects showing extremely large predictive uncertainties, is still present. The latter reinforces the interpretation that the effect is mainly related to the model and its uncertainty treatment, rather than to a fixed set of problematic stars. However, with $\log g$ included, the absolute residuals metrics have lower values than those of Section 5.2 (Table 4), consistently with the results of Moya & López-Sastre (2022).

Nevertheless, the signed biases are slightly amplified. This is explained by Figure 4, where the asteroseismic stars are concentrated in a relatively narrow mass range, approximately between $0.8\,M_\odot$ and $1.5\,M_\odot$. Since the holdout test is evaluated against asteroseismic stars from the same population, which mostly occupy the same region of the HR diagram, MARD consequently decreases. However, the predictive performance is expected to be less reliable for stars outside this range, where systematic trends are reflected in the larger absolute MRD values.

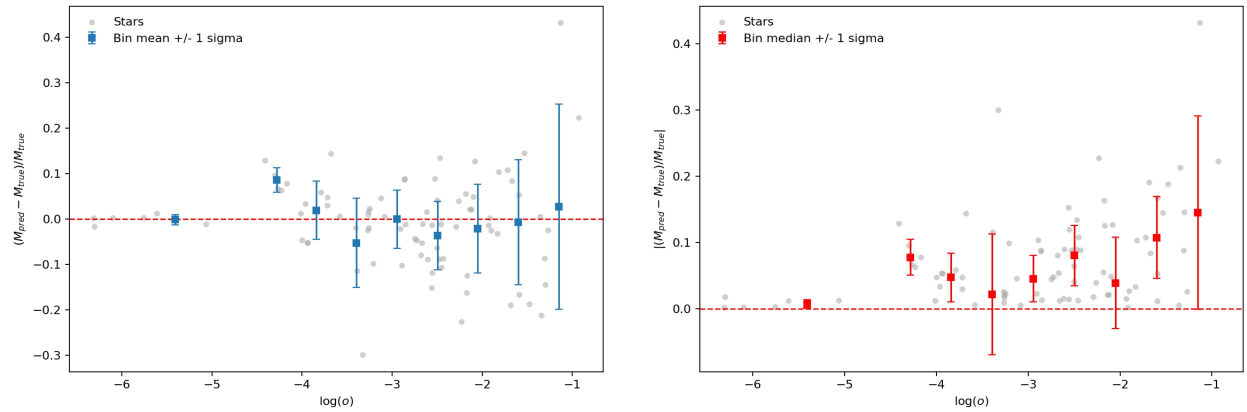
Regarding the stacking model, BHS yields the best or nearly best predictions across the evaluated metrics. Although HBNN has a slightly lower MRD, this may reflect the cancellation of systematic overpredictions and underpredictions, as discussed in previous sections. BHS instead provides a more balanced compromise: its MARD is almost identical to that of GP, and its absolute MRD remains well below 1%.

Results: oblateness

The effect of binary interaction was studied by predicting the masses of the stars in the original dataset that belong to eclipsing-binary systems. From this group, only inputs with an available orbital semi-major axis a could be used, since this parameter is required to compute the oblateness. As explained in Section 4.2, this leaves 295 stars with available oblateness estimates before applying the final selection cuts. Once o was computed, the models trained on the asteroseismic stars were applied to these eclipsing-binary components, and the resulting mass residuals were analysed as a function of oblateness.

To avoid interpreting extrapolation effects as physical binary effects, the eclipsing-binary sample was restricted to stars well covered by the asteroseismic training set. First, only stars with $0.950\,M_\odot < M < 1.500\,M_\odot$ were retained, corresponding approximately to the 5th and 95th mass percentiles of the asteroseismic sample. In addition, the final sample was restricted to the input-parameter domain $5500 < T_{\rm eff} < 6800\,\mathrm{K}$, $3.750 < \log g < 4.450$, $1.000 < L/L_\odot < 11.000$ and $-0.500 < [\mathrm{Fe/H}] < 0.270$, corresponding to the 2nd and 98th percentiles of the asteroseismic training distribution. The final eclipsing-binary sample contains $n = 85$ stars. This conservative selection reduces the number of systems, but makes the comparison more reliable, since the remaining stars lie inside the region of parameter space where the model is expected to have predictive validity.

The analysis was carried out using $\log o$, since the oblateness values span several orders of magnitude, as Figure 8 shows. Globally, the signed residuals show no strong systematic tendency towards either overprediction or underprediction. Pearson and Spearman tests support this interpretation, since both yield correlation coefficients close to zero. The mean fractional residual is $\overline{\delta M} = -0.85\%$, while the standard deviation reaches 10.43%, driven partly by a small number of large residuals and the reduced number of training stars. Therefore, the model is approximately unbiased on the selected eclipsing-binary sample, but the scatter is not negligible.



(a) Signed fractional mass residuals.

(b) Absolute fractional mass residuals.

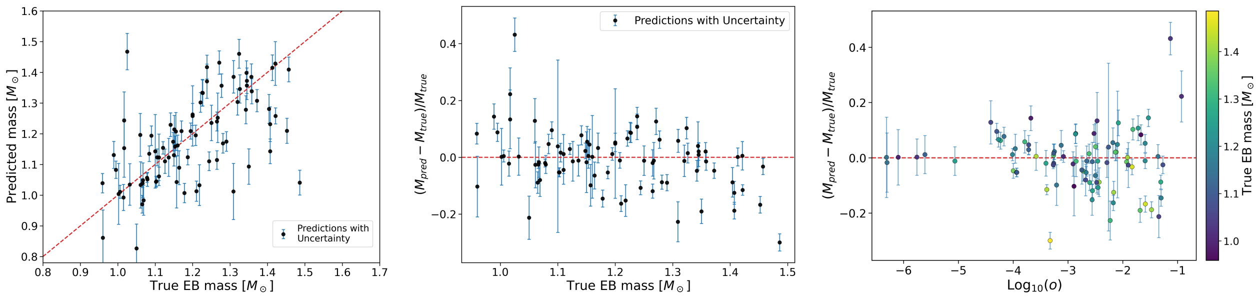
Figure 8: Dependence of the BHS mass residuals on stellar oblateness o, as defined in Equation 4, for the selected eclipsing-binary sample. The model uses T_{eff}, L, $\log g$ and [Fe/H] as input parameters. Grey points correspond to individual eclipsing-binary components, and the blue and red squares show binned means and medians with 1σ scatter. The sample contains $n = 85$ stars.

Figure 8b shows that the dispersion of the residuals increases with o. In the signed residual analysis, the standard deviation of δM rises from values of a few percent at low oblateness to much larger values in the most deformed systems: 1.06%, 2.65%, 6.50%, 9.89%, 6.23%, 7.57%, 9.77%, 13.87% and 22.74%. The overall behaviour is consistent with a degradation of precision at larger oblateness. If a residual scatter of about 10% is adopted as a practical tolerance for acceptable predictions, the transition occurs around $\log o \simeq -2$, corresponding to $o \simeq 0.01$. The same trend is visible in the absolute fractional residuals: most bins below this value remain below about 7%, whereas higher-oblateness bins reach approximately 15%. Therefore, the main effect of oblateness is not a clear change in the sign of the residuals, but an increase in their spread.

This distinction is important. The results do not indicate a systematic loss of accuracy, since the average residual remains close to zero. Instead, they indicate a loss of precision: as the stars become more oblate, the model predictions become more dispersed around the true eclipsing-binary masses. In this sample, $o \simeq 0.01$ provides an approximate practical limit beyond which the predictions begin to deviate more strongly from the behaviour expected for the approximately isolated stellar evolution represented by the asteroseismic training sample. Nevertheless, this interpretation should remain cautious, because the number of stars is limited and the inferred threshold depends on the adopted tolerance and the chosen bin edges.

Since Sections 5.1 and 5.2 show that stellar mass has a strong effect on the predictions, it is necessary to check whether the observed increase in scatter could instead be driven by M. Figure 9 shows that this is not the case: the residual dispersion does not display a clear dependence on stellar mass. If anything, the residual panel in Figure 9b shows a slight tendency for the models to pull predictions towards values close to $1.25\,M_{\odot}$. This behaviour was already discussed in detail and is not associated with an increase in prediction scatter; the precision loss appears to be mass-independent.

The mass-coloured residual panel in Figure 9c adds the true stellar mass as a colour scale. The points scatter around zero, and many error bars cross the zero-residual line. At the same time, the increase in dispersion at larger oblateness remains visible and not associated with increasing M_{true}, showing that the effect is not confined to a single mass interval.



(a) Predicted masses. **(b)** Fractional residuals. **(c)** Fractional residuals, colour-coded by true mass.

Figure 9: Analysis of BHS residuals for the selected eclipsing-binary sample of 85 stars. The model uses T_{eff}, L, $\log g$ and [Fe/H] as input parameters. The left panel compares predicted and reference EB masses, the central panel shows the corresponding fractional residuals as a function of true mass, and the right panel shows the fractional residuals as a function of stellar oblateness o, colour-coded by true mass. Error bars show the propagated predictive uncertainty, and the dashed red line marks zero residual.

To test the dependence of the residuals on the different input parameters, a multivariable linear regression was performed on the eclipsing-binary residuals. Each feature, together with o and M, was assigned a regression coefficient, β, quantifying its influence on the residual behaviour. Since the predictors were standardised, the coefficients can be directly compared.

The signed residuals are not significantly associated with oblateness, with $\beta_{\log o} = 0.002 \pm 0.005$, consistent with zero; o does not introduce an overall positive or negative bias in the predictions. Instead, the signed residuals are mainly associated with the stellar parameters themselves. As already seen in Figure 9, mass affects the tendency towards overprediction or underprediction, with $\beta_M = -0.108 \pm 0.005$. Other relevant contributions are found for T_{eff}, with $\beta_{T_{\mathrm{eff}}} = 0.077 \pm 0.007$, and for [Fe/H], with $\beta_{[\mathrm{Fe/H}]} = 0.048 \pm 0.002$.

The result changes when the absolute fractional residual is used as the response variable. In this case, $\log o$ is the only statistically significant predictor, with $\beta_{\log o} = 0.030 \pm 0.012$. Since the predictors are standardised, this means that a one-standard-deviation increase in $\log o$ is associated with an increase of about 3.0 percentage points in $|\delta M|$. The remaining predictors are not significant in this regression.

The uncertainty calibration also suggests that eclipsing-binary stars are harder to predict than stars drawn from the training distribution, since an additional source of scatter appears when the asteroseismic-trained model is applied to eclipsing binaries. Only 30/85 stars, or 35.3%, have their true mass within 1σ of the predictive interval. These results also qualify the uncertainty-aware nature of the framework: observational uncertainties are propagated into posterior predictive distributions, but the resulting intervals are not perfectly calibrated in all regimes.

6 Conclusions

This work employed an uncertainty-aware stacking approach to obtain posterior predictive distributions for stellar masses using probabilistic machine-learning models. Three level-0 models, BART, HBNN and GP, were combined through Bayesian Hierarchical Stacking. The training set was constructed from high-quality reference catalogues based mainly on asteroseismology, detached eclipsing binaries and interferometry. The starting point was the 726 main-sequence-star catalogue of Moya et al. (2018), which was extended with recent literature sources published from 2018 onwards. After cleaning and cross-matching the samples, the final usable set consisted of 767 main-sequence stars, all of which had M, T_{eff}, L, [Fe/H] and ρ; 729 also had $\log g$, and 295 had o.

The first experiment tested the models on FGK-type stars, since this was the dominant stellar population in the final sample. Using a 20% holdout set yielded mean absolute relative differences between predicted and reference masses which were all close to 5%, with the level-1 BHS model giving the best overall performance. HBNN and BART showed a systematic tendency to pull predictions towards approximately $1.25\,M_{\odot}$, where the training set is most densely populated. In contrast, GP showed no clear mass-dependent bias, although it produced a small number of outliers with unphysically large predictive uncertainties. The stacking model benefited from the complementary behaviour of the level-0 predictors and achieved an MRD of 0.29%, indicating negligible global bias.

The second experiment focused on a feature set with reduced model dependence, using T_{eff}, L and [Fe/H] to estimate stellar masses. The qualitative behaviour of the three level-0 models and the stacking model remained similar to that found in the previous experiment, although the predictive precision worsened by approximately two percentage points. The models also showed a small overpredicting tendency, with the BHS model giving an MRD of -1.36%. The predictions were then compared with evolutionary-grid estimates based on PARSEC (Bressan et al., 2012) and with linear-relation masses presented by Moya et al. (2018). The three approaches gave broadly compatible, but not identical results. A natural continuation of this analysis would be to combine all three approaches within a single stacking framework, so that the final estimate can benefit from the strengths of each method.

The third experiment showed that models trained on approximately single-star data can be transferred to eclipsing-binary components, but not without limitations. The oblateness parameter o was introduced to quantify the degree of interaction between the two components, and the models were trained only on the assembled asteroseismic sample. The differences between the predicted and reference masses for eclipsing-binary stars were then analysed as a function of o. For the selected sample of 85 eclipsing-binary stars, oblateness did not introduce a significant signed mass bias, with $\beta_{\log o} = 0.002 \pm 0.005$. However, it did increase the absolute residuals, with $\beta_{\log o} = 0.030 \pm 0.012$, indicating a loss of precision rather than a systematic loss of accuracy. A practical transition point was found around $o \sim 0.01$, above which the residual scatter becomes large enough for the predictions to be considered less reliable.

Conclusiones

Este trabajo tomó un enfoque de stacking con tratamiento explícito de incertidumbres para obtener distribuciones predictivas posteriores de masas estelares mediante modelos probabilísticos de machine learning. Tres modelos de nivel 0, BART, HBNN y GP, fueron combinados mediante Bayesian Hierarchical Stacking. El dataset de entrenamiento se construyó a partir de catálogos de referencia de alta calidad, basados principalmente en astrosismología, binarias eclipsantes e interferometría. El punto de partida fue el catálogo de 726 estrellas de la secuencia principal recopilado por Moya et al. (2018), que se amplió con fuentes bibliográficas recientes publicadas a partir de 2018. Tras la limpieza y el cruce de las muestras, el conjunto final utilizable quedó formado por 767 estrellas de la secuencia principal, todas ellas con M*,* T_{eff}*,* L*,* $[\mathrm{Fe/H}]$ *y* ρ*; 729 también tenían* $\log g$*, y 295 tenían* o*.*

El primer experimento evaluó los modelos en estrellas de tipo FGK, dado que esta era la población estelar dominante en la muestra final. Utilizando un conjunto de prueba reservado del 20%*, las diferencias relativas absolutas medias entre las masas predichas y las masas de referencia fueron todas próximas al* 5%*, con el modelo de stacking proporcionando el mejor rendimiento global. HBNN y BART mostraron una tendencia sistemática a desplazar las predicciones hacia aproximadamente* $1.25\,M_{\odot}$*, donde el conjunto de entrenamiento se muestra más densamente poblado. En cambio, GP no mostró un sesgo claro dependiente de la masa, aunque produjo un reducido número de predicciones con incertidumbres físicamente inverosímiles. El modelo de stacking se benefició del comportamiento complementario de los predictores de nivel 0 y alcanzó un MRD de* 0.29%*, lo que indica un sesgo global despreciable.*

El segundo experimento se centró en un conjunto de variables de entrada con menor dependencia de modelos, utilizando T_{eff}*,* L *y* $[\mathrm{Fe/H}]$*. El comportamiento cualitativo de los tres modelos de nivel 0 y del BHS se mantuvo similar al encontrado en el experimento anterior, aunque la precisión predictiva empeoró en aproximadamente dos puntos porcentuales. Los modelos también mostraron una leve tendencia a la sobrepredicción, con BHS dando un MRD de* -1.36%*. Las predicciones se compararon posteriormente con estimaciones basadas en modelos estelares de PARSEC (Bressan et al., 2012) y con predicciones mediante las relaciones lineales presentadas por Moya et al. (2018). Los tres enfoques dieron resultados compatibles, aunque no idénticos. La continuación natural de este análisis sería combinar los tres métodos dentro de un único marco de stacking, de modo que la estimación final pueda beneficiarse de las fortalezas de cada método.*

El tercer experimento mostró que los modelos entrenados con datos de estrellas individuales pueden transferirse a componentes de binarias eclipsantes, aunque no sin limitaciones. El parámetro de oblaticidad o *se introdujo para cuantificar el grado de interacción entre las dos componentes, y los modelos se entrenaron únicamente con la muestra astrosismológica recopilada. Las diferencias entre las predicciones de masas y los valores de referencia para las binarias eclipsantes se analizaron entonces como función de* o*. Para la muestra seleccionada, 85 estrellas binarias, la oblaticidad no introdujo un sesgo signado de masa significativo, con* $\beta_{\log o} = 0.002 \pm 0.005$*. Sin embargo, sí causó un aumento en los residuos absolutos, con* $\beta_{\log o} = 0.030 \pm 0.012$*, lo que indica una pérdida de precisión más que una pérdida sistemática de exactitud. Se encontró un punto de transición práctico alrededor de* $o \sim 0.01$*, por encima* *del cual la dispersión de los residuos se vuelve suficientemente grande como para considerar las predicciones menos fiables.*

References

Ariel Stellar Characterisation Working Group. n.d., Age Mass Radius sub-WG, `https://sites.google.com/inaf.it/arielstellarcatalogue/the-team/age-mass-radius-sub-wg`

Asplund, M., Grevesse, N., Sauval, A. J., & Scott, P. 2009, Annual Review of Astronomy and Astrophysics, 47, 481, doi: `10.1146/annurev.astro.46.060407.145222`

Bardenet, R., Doucet, A., & Holmes, C. 2017, Journal of Machine Learning Research, 18, 1. `https://jmlr.org/papers/v18/15-205.html`

Bellinger, E. P., Hekker, S., Angelou, G. C., Stokholm, A., & Basu, S. 2019, Astronomy & Astrophysics, 622, A130, doi: `10.1051/0004-6361/201834461`

Bilir, S., Karaali, S., Ak, S., et al. 2008, Monthly Notices of the Royal Astronomical Society, 390, 1569, doi: `10.1111/j.1365-2966.2008.13839.x`

Blundell, C., Cornebise, J., Kavukcuoglu, K., & Wierstra, D. 2015, in Proceedings of Machine Learning Research, Vol. 37, Proceedings of the 32nd International Conference on Machine Learning, ed. F. Bach & D. Blei (Lille, France: PMLR), 1613–1622. `https://arxiv.org/abs/1505.05424`

Bressan, A., Marigo, P., Girardi, L., et al. 2012, Monthly Notices of the Royal Astronomical Society, 427, 127, doi: `10.1111/j.1365-2966.2012.21948.x`

Chabrier, G., & Baraffe, I. 2000, Annual Review of Astronomy and Astrophysics, 38, 337, doi: `10.1146/annurev.astro.38.1.337`

Chandrasekhar, S. 1933, Monthly Notices of the Royal Astronomical Society, 93, 462, doi: `10.1093/mnras/93.6.462`

Chaplin, W. J., & Miglio, A. 2013, Annual Review of Astronomy and Astrophysics, 51, 353, doi: `10.1146/annurev-astro-082812-140938`

Chipman, H. A., George, E. I., & McCulloch, R. E. 2010, The Annals of Applied Statistics, 4, 266, doi: `10.1214/09-AOAS285`

Cho, S., Kim, D., & Hazlett, C. 2026, Political Analysis, 1, doi: `10.1017/pan.2026.10032`

Eker, Z., Bakış, V., Bilir, S., et al. 2018, Monthly Notices of the Royal Astronomical Society, 479, 5491, doi: `10.1093/mnras/sty1834`

Graczyk, D., Pietrzyński, G., Gałan, C., et al. 2021, Astronomy & Astrophysics, 649, A109, doi: `10.1051/0004-6361/202140571`

—. 2022, Astronomy & Astrophysics, 666, A128, doi: `10.1051/0004-6361/202244122`

Hastings, W. K. 1970, Biometrika, 57, 97, doi: `10.1093/biomet/57.1.97`

Hełminiak, K. G., Konacki, M., Maehara, H., et al. 2019, Monthly Notices of the Royal Astronomical Society, 484, 451, doi: `10.1093/mnras/sty3528`

Hill, J., Linero, A. R., & Murray, J. 2020, Annual Review of Statistics and Its Application, 7, 251, doi: `10.1146/annurev-statistics-031219-041110`

Izmailov, P., Vikram, S., Hoffman, M. D., & Wilson, A. G. G. 2021, in Proceedings of Machine Learning Research, Vol. 139, Proceedings of the 38th International Conference on Machine Learning, ed. M. Meila & T. Zhang (PMLR), 4629–4640. `https://proceedings.mlr.press/v139/izmailov21a.html`

Jospin, L. V., Buntine, W., Boussaid, F., Laga, H., & Bennamoun, M. 2022, IEEE Computational Intelligence Magazine, 17, 29, doi: `10.1109/MCI.2022.3155327`

Laughlin, G., Bodenheimer, P., & Adams, F. C. 1997, The Astrophysical Journal, 482, 420, doi: 10.1086/304125

Le, Q. V., Smola, A. J., & Canu, S. 2005, in Proceedings of the 22nd International Conference on Machine Learning (Association for Computing Machinery), 489–496, doi: 10.1145/1102351.1102413

Li, Y., & Chao, X. 2021, Frontiers in Plant Science, 12, 811241, doi: 10.3389/fpls.2021.811241

Ligi, R., Creevey, O., Mourard, D., et al. 2016, Astronomy & Astrophysics, 586, A94, doi: 10.1051/0004-6361/201527054

Maas, A. L., Hannun, A. Y., & Ng, A. Y. 2013, in Proceedings of the ICML 2013 Workshop on Deep Learning for Audio, Speech and Language Processing, 3. https://ai.stanford.edu/~amaas/papers/relu_hybrid_icml2013_final.pdf

Maxted, P. F. L. 2023, Monthly Notices of the Royal Astronomical Society, 522, 2683, doi: 10.1093/mnras/stad1112

McElreath, R. 2020, Statistical Rethinking: A Bayesian Course with Examples in R and Stan, 2nd edn. (Boca Raton: Chapman and Hall/CRC), doi: 10.1201/9780429029608

Moya, A., & López-Sastre, R. J. 2022, Astronomy & Astrophysics, 663, A112, doi: 10.1051/0004-6361/202142930

Moya, A., Zuccarino, F., Chaplin, W. J., & Davies, G. R. 2018, The Astrophysical Journal Supplement Series, 237, 21, doi: 10.3847/1538-4365/aacdae

Nsamba, B., Weiss, A., & Kamulali, J. 2025, Monthly Notices of the Royal Astronomical Society, 536, 2558, doi: 10.1093/mnras/stae2704

Pecaut, M. J., & Mamajek, E. E. 2013, The Astrophysical Journal Supplement Series, 208, 9, doi: 10.1088/0067-0049/208/1/9

Quiñonero-Candela, J., & Rasmussen, C. E. 2005, Journal of Machine Learning Research, 6, 1939

Rasmussen, C. E., & Williams, C. K. I. 2006, Gaussian Processes for Machine Learning (Cambridge, MA: MIT Press), doi: 10.7551/mitpress/3206.001.0001

Sayeed, M., Huber, D., Chontos, A., & Li, Y. 2025, The Astronomical Journal, 170, 212, doi: 10.3847/1538-3881/adf648

Serenelli, A., Weiss, A., Aerts, C., et al. 2021, Astronomy and Astrophysics Review, 29, 4, doi: 10.1007/s00159-021-00132-9

Soubiran, C., Creevey, O. L., Lagarde, N., et al. 2024, Astronomy & Astrophysics, 682, A145, doi: 10.1051/0004-6361/202347136

Southworth, J. 2015, in Astronomical Society of the Pacific Conference Series, Vol. 496, Living Together: Planets, Host Stars and Binaries, ed. S. M. Rucinski, G. Torres, & M. Zejda (Astronomical Society of the Pacific), 164–166. https://arxiv.org/abs/1411.1219

Tamames-Rodero, V., Moya, A., López, R. J., & Sarro, L. M. 2025, Unveiling the Power of Uncertainty: A Journey into Bayesian Neural Networks for Stellar Dating, doi: 10.48550/arXiv.2503.21153

Tsantaki, M., Andreasen, D. T., Teixeira, G. D. C., et al. 2018, Monthly Notices of the Royal Astronomical Society, 473, 5066, doi: 10.1093/mnras/stx2564

Wolpert, D. H. 1992, Neural Networks, 5, 241, doi: 10.1016/S0893-6080(05)80023-1

Xiong, J., Liu, C., Li, J., et al. 2023, The Astronomical Journal, 165, 30, doi: 10.3847/1538-3881/aca288

Yao, Y., Pirš, G., Vehtari, A., & Gelman, A. 2022, Bayesian Analysis, 17, 1043, doi: 10.1214/21-BA1287

Yıldız, M., Çelik Orhan, Z., & Kayhan, C. 2019, Monthly Notices of the Royal Astronomical Society, 489, 1753, doi: 10.1093/mnras/stz2223

Zhang, H., Nettleton, D., & Zhu, Z. 2019, Regression-Enhanced Random Forests, doi: 10.48550/arXiv.1904.10416